TABLEAU BASIC

Q1. Explain what Tableau is and why it is preferred over traditional spreadsheet tools for data analysis. Also mention any two industries where Tableau is commonly used and explain one use case

- Tableau is a data visualization and business intelligence (BI) tool used to analyze data and present it through interactive charts, graphs, maps, and dashboards. It helps businesses understand complex data and make better decisions.

Why is Tableau preferred over traditional spreadsheet tools?

1. Interactive dashboards: Users can filter data and explore different views easily.
2. Better visualization: It creates attractive charts, graphs, and maps to represent data clearly.
3. Handles large datasets: It can connect to and analyze large amounts of data from multiple sources.
4. Easy to use: Its drag-and-drop interface allows users to create visualizations without extensive coding.
5. Real-time insights: It can connect to supported live data sources to help users monitor changing information.
6. Data integration: It can combine data from sources such as Excel, databases, and cloud services.

Two industries where Tableau is commonly used

1. Healthcare industry

- Use case: Hospitals use Tableau to analyze patient records, monitor bed occupancy, and track treatment outcomes.
- Example: A hospital can create a dashboard showing the number of patients admitted each month to help manage staff and resources.

2. Retail industry

- Use case: Retail companies use Tableau to analyze sales performance, customer behavior, and product demand.
- Example: A retail store can create a dashboard showing sales by region and product to identify its best-selling products and improve inventory planning.

Q8. What is the difference between a Worksheet and a Dashboard in Tableau? Why is a dashboard more useful for decision-makers?

- A Worksheet in Tableau is used to create an individual visualization, such as a chart, graph, or table. A Dashboard combines multiple worksheets into a single screen to provide an overall view of important information.

Difference between Worksheet and Dashboard

Worksheet	Dashboard
Displays a single visualization or analysis.	Combines multiple visualizations in one view.
Used to analyze a specific aspect of data.	Used to monitor and compare different aspects of data.
Usually contains one chart, graph, or table.	Can contain charts, tables, filters, and other components.
Focuses on detailed analysis.	Focuses on overall business performance.
Example: A chart showing monthly sales.	Example: A screen showing monthly sales, regional sales, and profit.

Why is a Dashboard more useful for decision-makers?

1. Complete overview: It presents important business information in one place.
2. Saves time: Decision-makers do not need to open multiple worksheets separately.
3. Easy comparison: It helps compare sales, profit, and performance across regions or products.
4. Interactive analysis: Filters allow users to explore specific data according to their needs.
5. Better decision-making: Visual summaries help identify trends, problems, and opportunities quickly.

Example: A sales manager can use a Tableau dashboard to view total sales, monthly sales trends, regional performance, and profit together. This helps the manager identify weak-performing regions and take corrective action.

Q9. What is Tableau Prep, and why is it important in real-world data projects? What is Tableau Prep, and why is it important in real-world data projects? Then explain how you would: Remove null values Remove duplicate records Change incorrect data types (Answer in steps, no tool access required.)

- Tableau Prep is a data preparation tool used to clean, organize, combine, and transform raw data before analyzing it in Tableau. It helps make data accurate, consistent, and ready for visualization.

Importance of Tableau Prep in real-world data projects

1. Data cleaning: Removes unwanted values, errors, and inconsistencies.
2. Duplicate removal: Eliminates repeated records to improve data accuracy.
3. Data transformation: Changes data formats and structures as required.
4. Data integration: Combines data from different sources.
5. Time-saving: Automates repetitive data preparation tasks.
6. Improved analysis: Ensures that reports and dashboards are based on reliable data.

Steps to clean data using Tableau Prep

1. Remove null values

Null values are missing or empty values in a dataset.

Steps:

1. Open Tableau Prep Builder and connect to your dataset.
2. Add a Clean Step to the data flow.
3. Identify the field containing null values.
4. Click the drop-down menu for that field.
5. Select the option to filter out null values.
6. Review the resulting data to ensure the necessary records remain.

Note: Depending on the project, missing values can also be replaced with appropriate values instead of deleting entire records.

2. Remove duplicate records

Duplicate records are repeated rows that may lead to incorrect calculations.

Steps:

1. Open the dataset in Tableau Prep.
2. Add a Clean Step.
3. Select the fields that define a unique record.
4. Use the Group and Replace or aggregation features where appropriate, or use a cleaning method that identifies duplicate rows.
5. Retain one record for each unique combination of the selected fields.

6. Check the output to confirm that duplicates have been handled correctly.

3. Change incorrect data types

Data types specify whether a field contains text, numbers, dates, or other kinds of data.

Steps:

1. Open the dataset in Tableau Prep.
2. Locate the field with the incorrect data type.
3. Click the data type icon next to the field name.
4. Select the correct type, such as String, Number, Date, or Date & Time.
5. Check whether all values convert correctly.
6. Resolve any conversion errors and verify the cleaned data.

Example: If an `Order Date` column is stored as text, change its data type to Date so Tableau can correctly analyze sales by month or year.

Q10. A retail manager wants answers to the following: Which category generates the highest profit? Which region performs poorly? How does sales change over time? a) Mention which charts you would use b) Identify one key insight that management could derive c) Explain how this helps in business decision-making

- A retail manager can use Tableau charts and dashboards to analyze profit by category, compare regional performance, and track sales over time.

a) Which charts would you use?

Business Question	Recommended Chart	Purpose
Which category generates the highest profit?	Bar Chart	Compares profit across product categories and identifies the most profitable category.
Which region performs poorly?	Bar Chart	Compares sales or profit across regions to identify underperforming regions.
How do sales change over time?	Line Chart	Shows sales trends by month, quarter, or year.

b) Identify one key insight for management

Suppose the Tableau dashboard reveals that:

- The Technology category generates the highest profit.
- The Central region has the lowest profit.
- Sales increase during the festive season and decline in certain months.

Key insight: Management should investigate the reasons for low profitability in the Central region while maintaining strong performance in the Technology category.

Note: These are hypothetical findings; actual insights depend on the retail dataset.

c) How does this help in business decision-making?

1. Improve profitability: Management can focus on high-profit categories and promote successful products.
2. Improve regional performance: The company can investigate low sales, high costs, or weak demand in underperforming regions.
3. Plan inventory: Sales trends help predict demand and avoid overstocking or stock shortages.
4. Optimize marketing: The business can run targeted promotions during periods of low sales.
5. Allocate resources: Management can direct budgets, staff, and inventory toward areas with the greatest potential.